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(54) **PARALLEL APPROACH TO DYNAMIC MESH ALIGNMENT**

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G06T 9/00 (2006.01)

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CPC **G06T 19/20** (2013.01); **G06T 9/001** (2013.01); **G06T 2210/52** (2013.01); **G06T 2219/2004** (2013.01)

(58) **Field of Classification Search**
CPC **G06T 19/20**; **G06T 2219/2004**; **G06T 2210/52**; **G06T 9/001**

See application file for complete search history.

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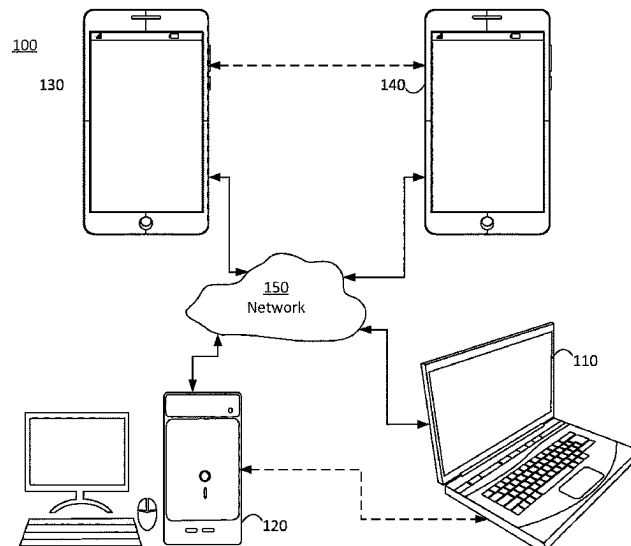
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(57) **ABSTRACT**

Method, apparatus, and system for parallel dynamic mesh alignment are provided. The process may include determining that a temporal alignment is present between one or more frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object, then, determining an intra-frame alignment scheme to spatially align charts within a frame. The process may also include applying the intra-frame alignment scheme to one or more corresponding charts in the one or more frames for inter-frame alignment; and processing the one or more frames in parallel based on the intra-frame alignment scheme and inter-frame alignment.

17 Claims, 7 Drawing Sheets



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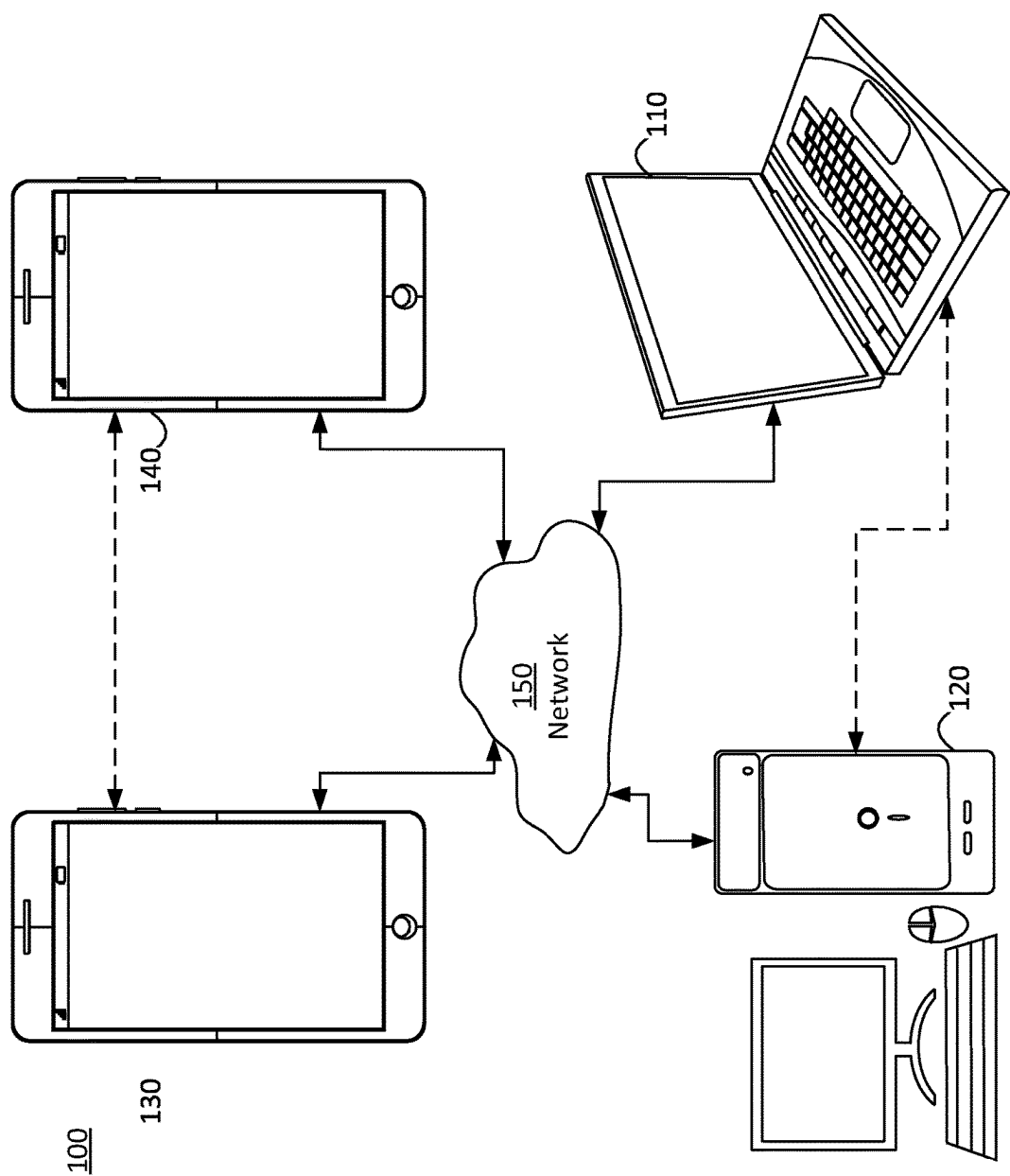


FIG. 1

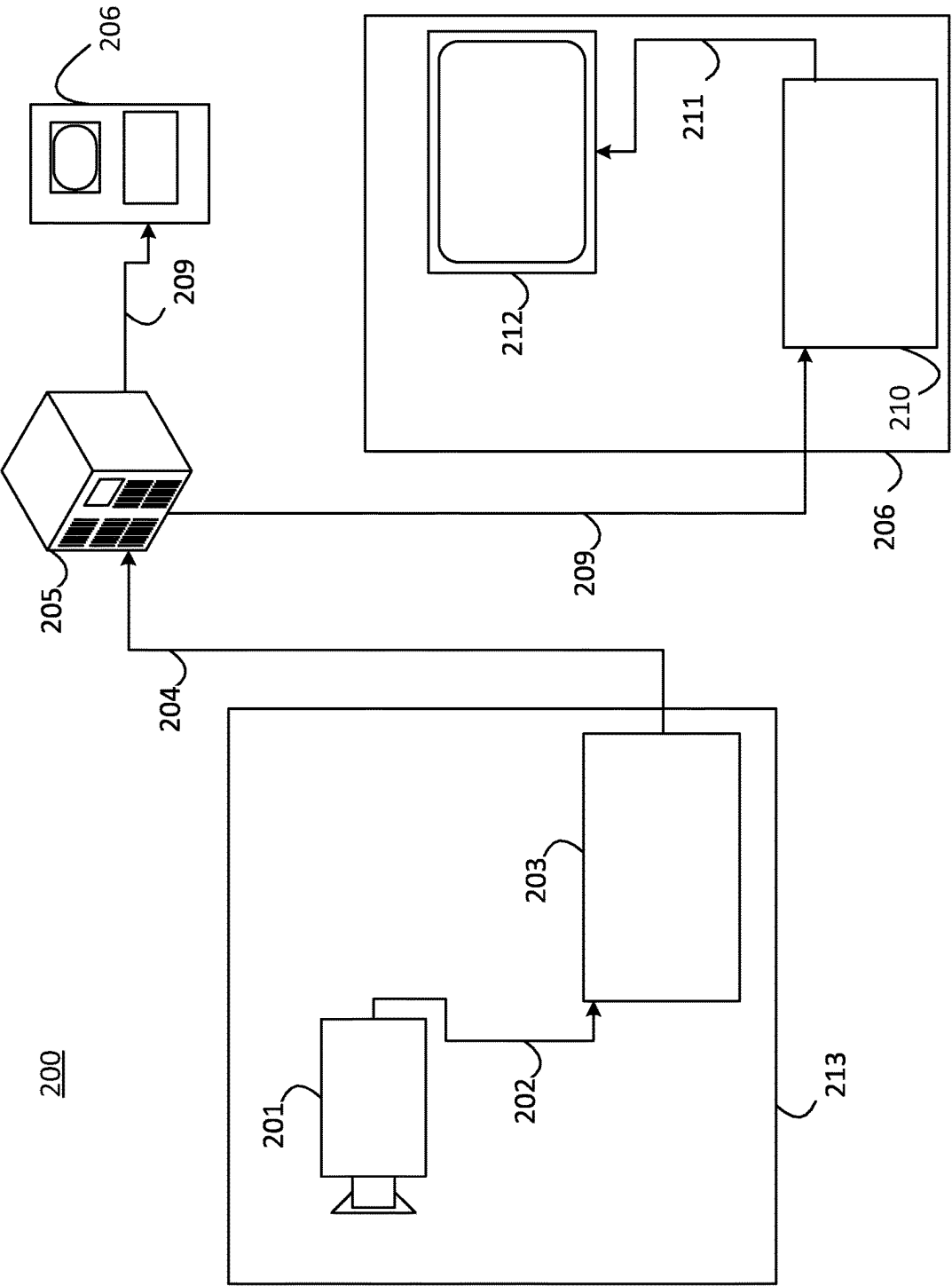


FIG. 2

FIG. 3

300

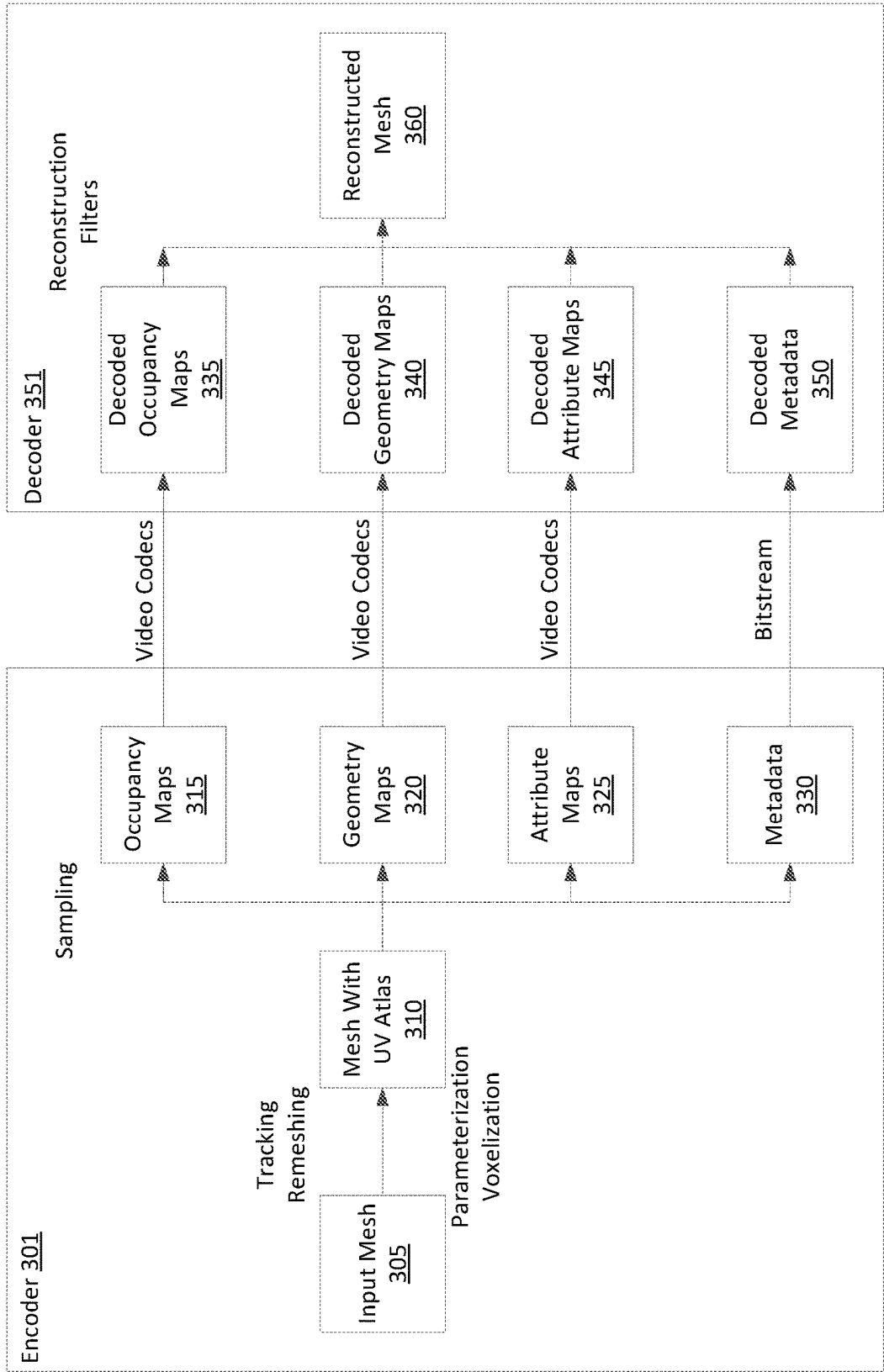


FIG. 4

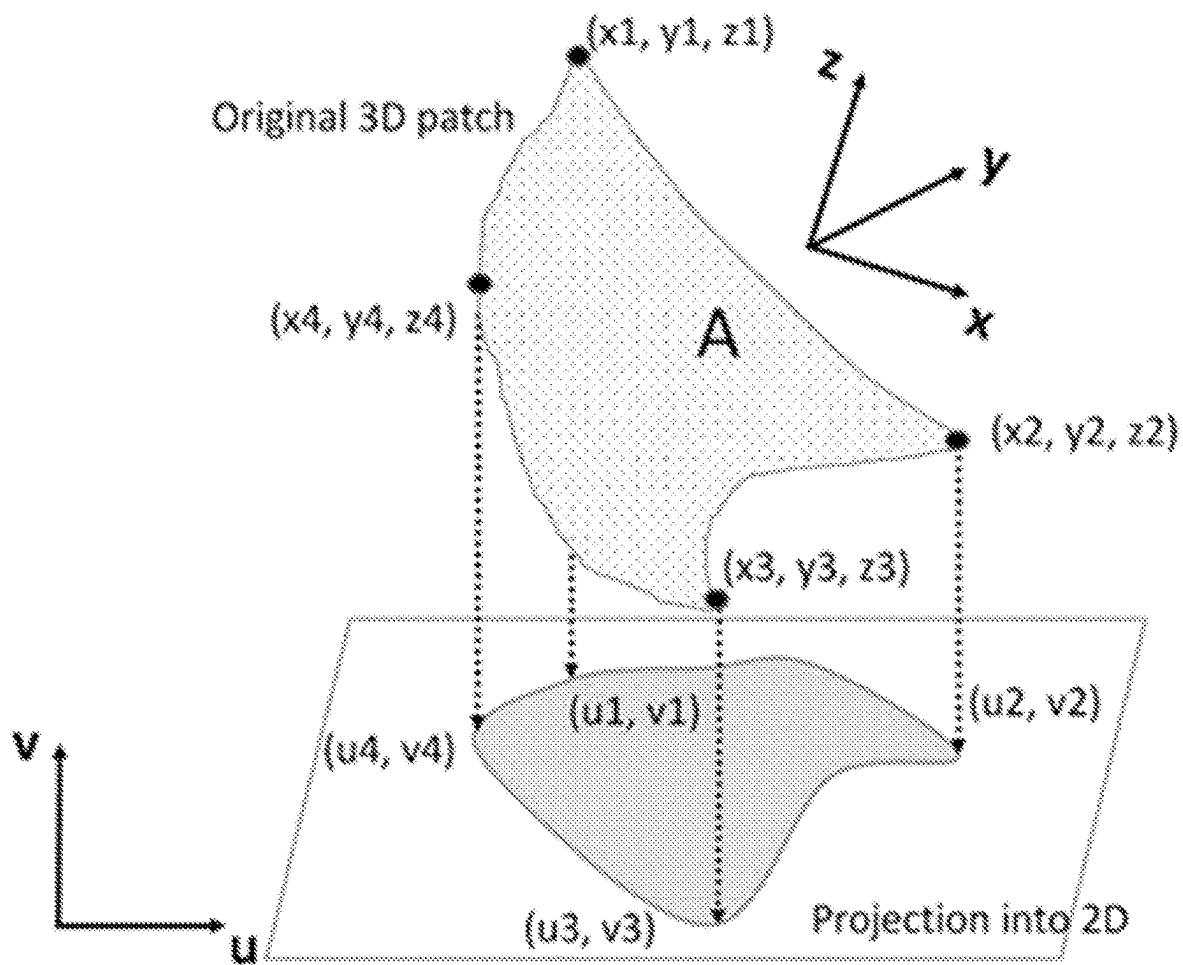
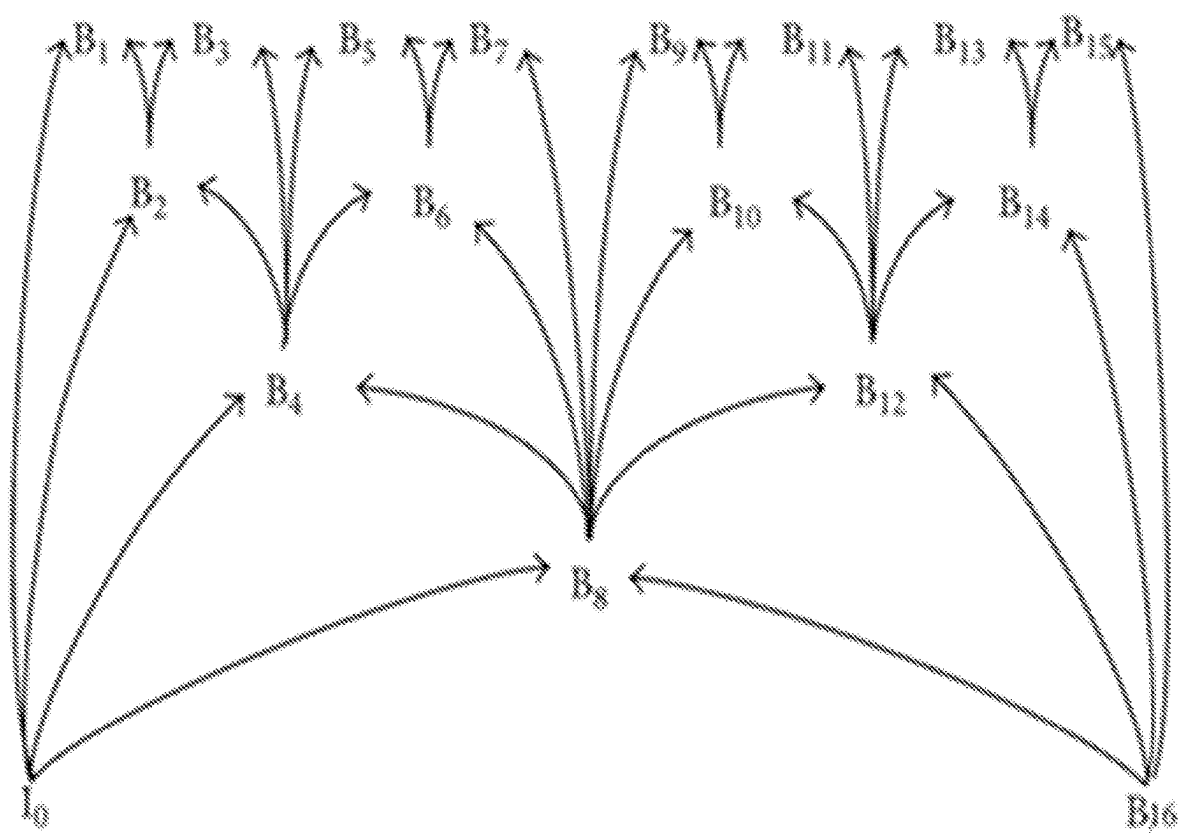
400

FIG. 5

500

600

FIG. 6

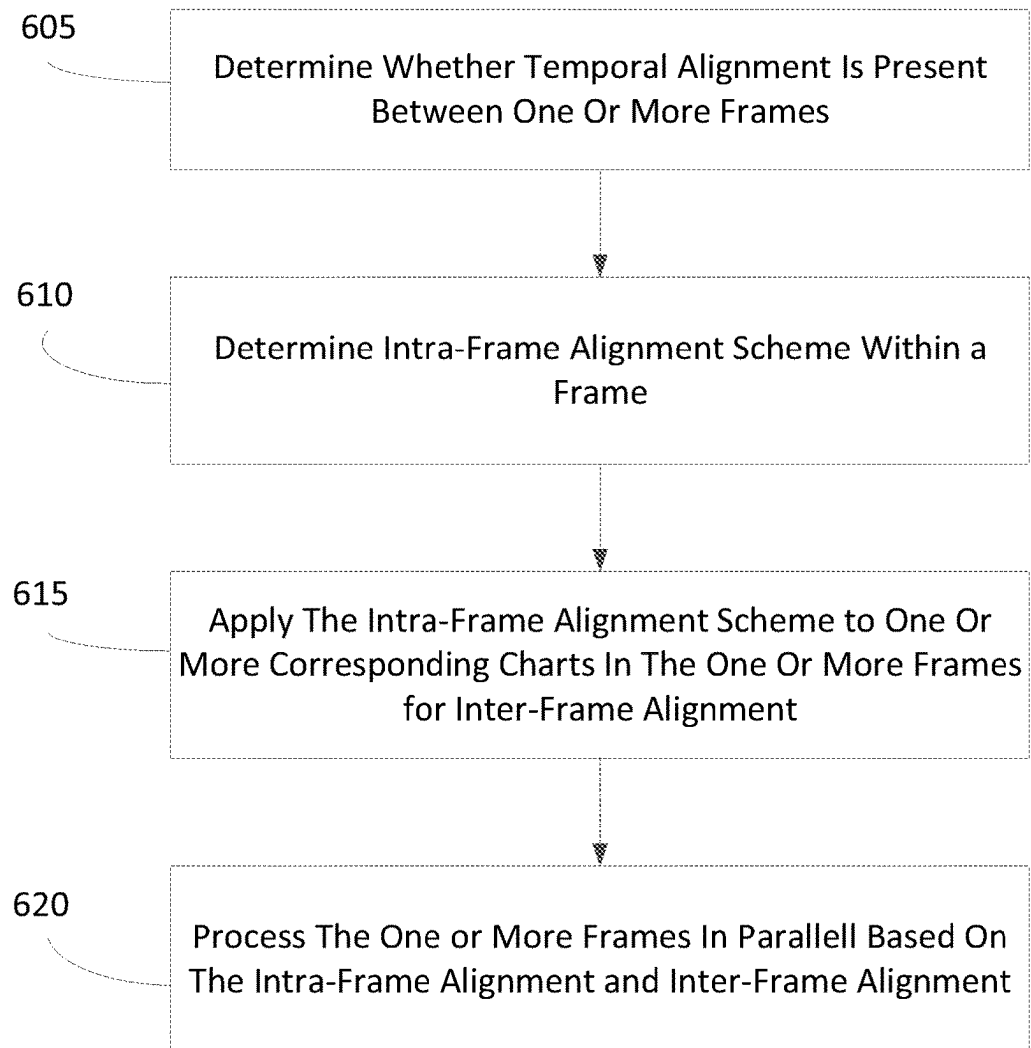
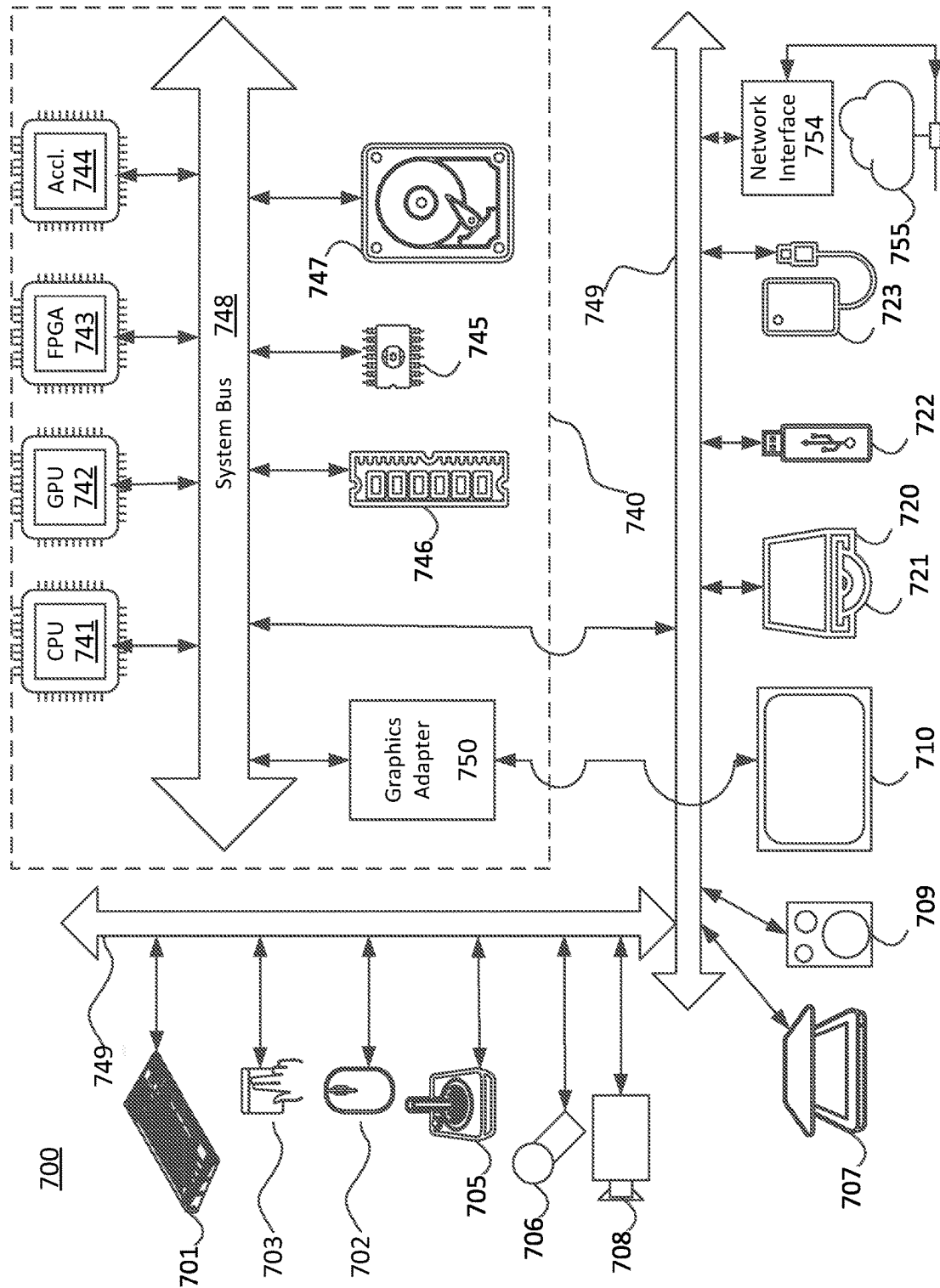


FIG. 7



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PARALLEL APPROACH TO DYNAMIC MESH ALIGNMENT

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from U.S. Provisional Application No. 63/323,886, filed on Mar. 25, 2022, the disclosure of which is incorporated herein by reference in its entirety.

FIELD

This disclosure is directed to a set of advanced video coding technologies. More specifically, the present disclosure is directed to video based dynamic mesh alignment and compression.

BACKGROUND

Advanced three-dimensional (3D) representations of the world are enabling more immersive forms of interaction and communication. To achieve realism in 3D representations, 3D models are becoming ever more sophisticated, and a significant amount of data is linked to the creation and consumption of these 3D models. 3D meshes are widely used to 3D model immersive content.

A 3D mesh may include several polygons that describe the surface of a volumetric object. A dynamic mesh sequence may require a large amount of data since it may have a significant amount of information changing over time. Therefore, efficient compression technologies are required to store and transmit such contents.

While mesh compression standards IC, MESHGRID, FAMC were previously developed to address dynamic meshes with constant connectivity and time varying geometry and vertex attributes. However, these standards do not take into account time varying attribute maps and connectivity information.

Furthermore, it is also challenging for volumetric acquisition techniques to generate a constant connectivity dynamic mesh, especially under real time constraints. This type of dynamic mesh content is not supported by the existing standards.

In addition to the above, many alignment methods, especially the temporal alignment, are based on sequential techniques. In other words, the processing of one frame depends on other frames, such as using the results of chart allocation from a previous frame as the basis for allocating the charts of the current frame. The dependency makes those methods unsuitable for parallelization hence are slow to encode/decode meshes. Therefore, methods eliminating this dependency are needed to enable parallelization and improve encoding and decoding efficiency.

SUMMARY

According to embodiments, a method for parallel dynamic mesh alignment may be provided. The method may include determining that a temporal alignment is present between one or more frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object; based on determining that the temporal alignment between the one or more frames is present, determining an intra-frame alignment scheme to spatially align charts within a frame; applying the intra-frame alignment scheme to one or more corresponding charts in the one

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or more frames for inter-frame alignment; and processing the one or more frames in parallel based on the intra-frame alignment scheme and the inter-frame alignment.

According to embodiments, an apparatus for parallel dynamic mesh alignment may be provided. The apparatus may include at least one memory configured to store program code; and at least one processor configured to read the program code and operate as instructed by the program code. The program code may include first determining code configured to cause the at least one processor to determine that a temporal alignment is present between one or more frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object; based on determining that the temporal alignment between the one or more frames is present, second determining code configured to cause the at least one processor to determine an intra-frame alignment scheme to spatially align charts within a frame; first applying code configured to cause the at least one processor to apply the intra-frame alignment scheme to one or more corresponding charts in the one or more frames for inter-frame alignment; and first processing code configured to cause the at least one processor to process the one or more frames in parallel based on the intra-frame alignment scheme and the inter-frame alignment.

According to embodiments, a non-transitory computer-readable medium stores computer instructions may be provided. The instructions may include one or more instructions that, when executed by one or more processors of a device for parallel dynamic mesh alignment, cause the one or more processors to determine that a temporal alignment is present between one or more frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object; based on determining that the temporal alignment between the one or more frames is present, determine an intra-frame alignment scheme to spatially align charts within a frame; apply the intra-frame alignment scheme to one or more corresponding charts in the one or more frames for inter-frame alignment; and process the one or more frames in parallel based on the intra-frame alignment scheme and the inter-frame alignment.

BRIEF DESCRIPTION OF THE DRAWINGS

Further features, the nature, and various advantages of the disclosed subject matter will be more apparent from the following detailed description and the accompanying drawings in which:

FIG. 1 is a schematic illustration of a simplified block diagram of a communication system, in accordance with embodiments of the present disclosure.

FIG. 2 is a schematic illustration of a simplified block diagram of a streaming system, in accordance with embodiments of the present disclosure.

FIG. 3 is a schematic illustration of a simplified block diagram of a video encoder and decoder, in accordance with embodiments of the present disclosure.

FIG. 4 is an exemplary illustration of projecting a 3D chart onto a 2D UV plane, in accordance with embodiments of the present disclosure.

FIG. 5 is an exemplary hierarchical grouping of frames or subgrouping of frames, in accordance with embodiments of the present disclosure.

FIG. 6 is an exemplary flow diagram illustrating parallel dynamic mesh alignment, in accordance with embodiments of the present disclosure.

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FIG. 7 is a diagram of a computer system suitable for implementing embodiments.

DETAILED DESCRIPTION

A mesh may include several polygons that describe the surface of a volumetric object. Its vertices in 3D space and the information of how the vertices are connected may define each polygon, referred to as connectivity information. Optionally, vertex attributes, such as colors, normals, etc., may be associated with the mesh vertices. Attributes may also be associated with the surface of the mesh by exploiting mapping information that parameterizes the mesh with 2D attribute maps. Such mapping may be defined using a set of parametric coordinates, referred to as UV coordinates or texture coordinates, and associated with the mesh vertices. 2D attribute maps may be used to store high resolution attribute information such as texture, normals, displacements etc. The high resolution attribute information may be used for various purposes such as texture mapping and shading.

As stated above, a 3D mesh or dynamic meshes may require a large amount of data since it may consist of a significant amount of information changing over time. Existing standards do not take into account time varying attribute maps and connectivity information. Existing standards also do not support volumetric acquisition techniques that generate a constant connectivity dynamic mesh, especially under real-time conditions.

Therefore, new mesh compression standard to directly handle dynamic meshes with time varying connectivity information and optionally time varying attribute maps is needed. Embodiments of the present disclosure enable efficient compression technologies to store and transmit such dynamic meshes. Embodiments of the present disclosure enable lossy and/or lossless compression for various applications, such as real-time communications, storage, free viewpoint video, AR and VR.

To achieve efficient compression, embodiments of the present disclosure employ spatial and temporal alignments to obtain intra-frame and/or inter-frame correlations. In related art, many of the alignment methods, especially the temporal alignment, are based on sequential techniques. In other words, the processing of one frame depends on other frames, such as using the results of chart allocation from a previous frame as the basis for allocating the charts of the current frame. The dependency makes those methods unsuitable for parallelization hence are slow to encode/decode meshes. Therefore, methods eliminating this dependency are needed to enable parallelization and improve encoding and decoding efficiency.

According to an aspect of the present disclosure, methods, systems, and non-transitory storage mediums for parallel processing of dynamic mesh compression are provided. Embodiments of the present disclosure may also be applied to static meshes.

With reference to FIGS. 1-2, an embodiment of the present disclosure for implementing encoding and decoding structures of the present disclosure are described.

FIG. 1 illustrates a simplified block diagram of a communication system 100 according to an embodiment of the present disclosure. The system 100 may include at least two terminals 110, 120 interconnected via a network 150. For unidirectional transmission of data, a first terminal 110 may code video data, which may include mesh data, at a local location for transmission to the other terminal 120 via the network 150. The second terminal 120 may receive the

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coded video data of the other terminal from the network 150, decode the coded data and display the recovered video data. Unidirectional data transmission may be common in media serving applications and the like.

FIG. 1 illustrates a second pair of terminals 130, 140 provided to support bidirectional transmission of coded video that may occur, for example, during videoconferencing. For bidirectional transmission of data, each terminal 130, 140 may code video data captured at a local location for transmission to the other terminal via the network 150. Each terminal 130, 140 also may receive the coded video data transmitted by the other terminal, may decode the coded data and may display the recovered video data at a local display device.

In FIG. 1, the terminals 110-140 may be, for example, servers, personal computers, and smart phones, and/or any other type of terminals. For example, the terminals (110-140) may be laptop computers, tablet computers, media players and/or dedicated video conferencing equipment. The network 150 represents any number of networks that convey coded video data among the terminals 110-140 including, for example, wireline and/or wireless communication networks. The communication network 150 may exchange data in circuit-switched and/or packet-switched channels. Representative networks include telecommunications networks, local area networks, wide area networks, and/or the Internet. For the purposes of the present discussion, the architecture and topology of the network 150 may be immaterial to the operation of the present disclosure unless explained herein below.

FIG. 2 illustrates, as an example of an application for the disclosed subject matter, a placement of a video encoder and decoder in a streaming environment. The disclosed subject matter can be used with other video enabled applications, including, for example, video conferencing, digital TV, storing of compressed video on digital media including CD, DVD, memory stick and the like, and so on.

As illustrated in FIG. 2, a streaming system 200 may include a capture subsystem 213 that includes a video source 201 and an encoder 203. The streaming system 200 may further include at least one streaming server 205 and/or at least one streaming client 206.

The video source 201 can create, for example, a stream 202 that includes a 3D mesh and metadata associated with the 3D mesh. The video source 201 may include, for example, 3D sensors (e.g. depth sensors) or 3D imaging technology (e.g. digital camera(s)), and a computing device that is configured to generate the 3D mesh using the data received from the 3D sensors or the 3D imaging technology. The sample stream 202, which may have a high data volume when compared to encoded video bitstreams, can be processed by the encoder 203 coupled to the video source 201. The encoder 203 can include hardware, software, or a combination thereof to enable or implement aspects of the disclosed subject matter as described in more detail below. The encoder 203 may also generate an encoded video bitstream 204. The encoded video bitstream 204, which may have a lower data volume when compared to the uncompressed stream 202, can be stored on a streaming server 205 for future use. One or more streaming clients 206 can access the streaming server 205 to retrieve video bit streams 209 that may be copies of the encoded video bitstream 204.

The streaming clients 206 can include a video decoder 210 and a display 212. The video decoder 210 can, for example, decode video bitstream 209, which is an incoming copy of the encoded video bitstream 204, and create an outgoing video sample stream 211 that can be rendered on

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the display **212** or another rendering device (not depicted). In some streaming systems, the video bitstreams **204**, **209** can be encoded according to certain video coding/compression standards.

FIG. **3** is an exemplary diagram of framework **300** for dynamic mesh compression and mesh reconstruction using encoders and decoders.

As seen in FIG. **3**, framework **300** may include an encoder **301** and a decoder **351**. The encoder **301** may include one or more input mesh **305**, one or more mesh with UV atlas **310**, occupancy maps **315**, geometry maps **320**, attribute maps **325**, and metadata **330**. The decoder **351** may include decoded occupancy maps **335**, decoded geometry maps **340**, decoded attribute maps **345**, decoded metadata **350**, and reconstructed mesh **360**.

According to an aspect of the present disclosure, the input mesh **305** may include one or more frames, and each of the one or more frames may be preprocessed by a series of operations and used to generate the mesh with UV atlas **310**. As an example, the preprocessing operations may include and may not be limited to tracking, parameterization, remeshing, voxelization, etc. In some embodiments, the preprocessing operations may be performed only on the encoder side and not the decoder side.

The mesh with UV atlas **310** may be a 2D mesh. The 2D mesh with UV atlas may be a mesh in which each vertex of the mesh may be associated with UV coordinates on a 2D atlas. The mesh with the UV atlas **310** may be processed and converted into a plurality of maps based on sampling. As an example, the UV atlas **310** may be processed and converted into occupancy maps, geometry maps, and attribute maps based on sampling the 2D mesh with UV atlas. The generated occupancy maps **335**, geometry maps **340**, and attribute maps **345** may be encoded using appropriate codecs (e.g., HVEC, VVC, AV1, etc.) and transmitted to a decoder. In some embodiments, metadata (e.g., connectivity information etc.) may also be transmitted to the decoder.

According to an aspect, the decoder **351** may receive the encoded occupancy maps, geometry maps, and attribute maps from an encoder. The decoder **351** may use appropriate techniques and methods, in addition to embodiments described herein, to decode the occupancy maps, geometry maps, and attribute maps. In an embodiment, decoder **351** may generate decoded occupancy maps **335**, decoded geometry maps **340**, decoded attribute maps **345**, and decoded metadata **350**. The input mesh **305** may be reconstructed into reconstructed mesh **360** based on the decoded occupancy maps **335**, decoded geometry maps **340**, decoded attribute maps **345**, and decoded metadata **350** using one or more reconstruction filters and techniques. In some embodiments, the metadata **330** may be directly transmitted to decoder **351** and the decoder **351** may use the metadata to generate the reconstructed mesh **360** based on the decoded occupancy maps **335**, decoded geometry maps **340**, and decoded attribute maps **345**. Post-filtering techniques, including but not limited to remeshing, parameterization, tracking, voxelization, etc., may also be applied on the reconstructed mesh **360**.

The input meshes with 2D UV atlases may have vertices, where each vertex of the mesh may have an associated UV coordinates on the 2D atlas. The occupancy, geometry, and attribute maps may be generated by sampling one or more points/positions on the UV atlas. Each sample position, if it is inside a polygon defined by the mesh vertices, may be occupied or unoccupied. For each occupied sample, one can

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calculate its corresponding 3D geometry coordinates and attributes by interpolating from the associated polygon vertices.

According to an aspect of the present disclosure, the sampling rate may be consistent over the whole 2D atlas. In some embodiments, the sampling rate for u and v axes may be different, making anisotropic remeshing possible. In some embodiments, the whole 2D atlas may be divided into multiple regions, such as slices or tiles, and each such region may have a different sampling rate.

According to an aspect of the present disclosure, the sampling rate for each region (or the entire 2D atlas) may be signaled in a high-level syntax, including but not limited to sequence header, frame header, slice header, etc. In some embodiments, sampling rate for each region (or the entire 2D atlas) may be chosen from a pre-established set of rates that have been assumed by both the encoder and decoder. Because the pre-established set of rates that are known by both the encoder and decoder, signaling of one particular sampling rate would require only signaling the index in the pre-established rate set. An example of such a pre-established set may be every 2 pixels, every 4 pixels, every 8 pixels, etc. In some embodiments, the sampling rate for each region (or the entire 2D atlas) of a mesh frame may be predicted from a pre-established rate set, from a previously used sampling rate in other already coded regions of the same frame, or from a previously used sampling rate in other already coded mesh frames.

In some embodiments, the sampling rate for each region (or the entire 2D atlas) may be based on some characteristic of each region (or the entire 2D atlas). As an example, the sample rate can be based on activity—for a rich-textured region (or the entire 2D atlas), or a region (or the entire 2D atlas) with high activity, the sample rate could be set higher. As another example, for a smooth region (or the entire 2D atlas), or a region (or the entire 2D atlas) with low activity, the sample rate could be set lower.

In some embodiments, the sampling rate for each region (or the entire 2D atlas) of a mesh frame may be signaled in a way that combination of prediction and direct signaling may be allowed. The syntax may be structured to indicate if a sampling rate will be predicted or directly signaled. When predicted, which of the predictor-sampling rate to be used may be further signaled. When directly signaled, the syntax to represent the value of the rate may be signaled.

FIG. **4** is an exemplary diagram **400** illustrating projecting a 3D chart onto a 2D UV plane (2D UV chart), in accordance with embodiments of the present disclosure.

As seen in FIG. **4**, a 3D mesh may be split into several charts, each of which may be projected on to a 2D plane. On the 2D UV plane, the attribute information of the mesh, such as texture information, can find it correspondence to the 3D vertices.

As seen in FIG. **4**, the UV coordinate information may be used to find the texture information of a 3D location (such as a vertex) on the 2D UV plane. It must be noted that, a person skilled in the art would know that for different mesh frames, the partitioning process of charts may not be the same, resulting various shapes of charts and allocations on the 2D planes across different mesh frames.

As stated above, a dynamic mesh sequence may require a large amount of data since it may consist of a significant amount of information changing over time. To achieve efficient compression, embodiments of the present disclosure employ spatial and temporal alignments to obtain intra-frame and/or inter-frame correlations. In related art, many of the alignment methods, especially the temporal

alignment, are based on sequential techniques. In other words, the processing of one frame depends on other frames, such as using the results of chart allocation from a previous frame as the basis for allocating the charts of the current frame. The dependency makes those methods unsuitable for parallelization hence are slow to encode/decode meshes. Therefore, methods eliminating this dependency are needed to enable parallelization and improve encoding and decoding efficiency.

Embodiments of the present disclosure are directed to aligning mesh texture maps and/or geometry images with different levels of parallelization.

In an embodiment, maximum temporal parallelization of alignment is achieved. In this embodiment, each frame may be processed independently while the temporal alignment between frames is maintained. In this embodiment, an intra-frame alignment scheme may be determined to spatially align charts within a frame. The intra-frame alignment may be applied to every frame. Given consecutive frames being similar, the resulting aligned charts from those frames are similar as well. Thus, both intra-frame and inter-frame alignments may be achieved and the frames may be processed independently and in parallel. In some embodiments, intra-frame alignment refers to charts in a neighboring location within the frame having similar property such as color values, etc. In some embodiments, inter-frame alignment refers to charts in a similar location across different frames having similar property such as color values, shapes, etc.

In some embodiments, the geometry information and/or texture information of the charts may be used to design the intra-frame alignment scheme.

When spatially aligning charts in geometry images, charts may be spatially aligned using any information of the charts, such as the 3D location, size, normal, and shape of the charts etc. As an example, the location of a chart may be represented by its centroid, which may be computed by averaging the 3D coordinates of all vertices in the chart or be approximated by averaging the 3D coordinates of the chart boundary vertices only. Then, the charts may be ordered and rasterized according to their locations by using the Morton code/Z-order of their centroid coordinates.

In addition to geometry information described hereinabove, texture information, such as the color or histogram of the charts, may also be taken into consideration when spatially aligning charts, when the texture maps are available.

When spatially aligning charts in texture maps, both geometry information and texture information may be used. The rationale of using geometry information for aligning texture charts is that, if the geometric changes between consecutive frames are small, the texture of geometrically nearby charts from those frames will be similar. For the texture information, beside the histogram mentioned above, the mean RGB/YUV pixel value of a chart that represents its color information may be used. In some embodiments, the mean RGB/YUV pixel value of a chart may be computed by averaging all pixel values in the chart or be approximated by only averaging those pixel values on the chart boundaries. Then, the charts may be ordered and rasterized according to the Morton code/Z-order of their mean RGB/YUV pixel values. In some embodiments, the charts may be sorted and rasterized according to other ordering of the mean RGB/YUV pixel values, such as using the corresponding locations in the visible spectrum.

In some embodiments, temporal alignment algorithms for placing charts on to 2D images, either for geometry infor-

mation or texture information, may permit the reference to a previous frames' results. However certain limitations may be imposed on such reference so that parallel processing is still possible to some extent. In one embodiment, the frames may be divided into subgroups that may be processed in parallel.

FIG. 5 illustrates an exemplary hierarchical coding structure with several layers or levels indicating dependency between groups or subgroups for limited parallel processing of frames, in accordance with embodiments of the present disclosure.

As seen in FIG. 5, structure 500 includes I0 and B16 at a lowest level while B1, B3, B5, B7, B9, B11, B13, B15 are at the highest level. In an embodiment, higher level frames may only refer to lower level frames for temporal alignment results, but not to refer to frames of the same level or higher levels. As an example, B8 may refer to I0's temporal alignment results. Thus, some parallel processing is possible for frames at the same level.

In some embodiments, the mesh frames may be partitioned into independent group of pictures (GoP) (e.g., B1, B2, . . . , B16, I0), so that each GoP may be processed in a parallel manner. Within a GoP, the above mentioned prediction structures (either hierarchical or linear) (e.g., structure 500) may be used.

As another embodiment, a first mesh frame may be processed independently, and all the other frames may be processed based on the first frame. Therefore, only one-frame latency may be introduced.

FIG. 6 is a flow diagram illustrating a process 600 for parallel dynamic mesh alignment, in accordance with embodiments of the present disclosure.

At operation 605, a temporal alignment is present between one or more frames may be determined. In some embodiments, the temporal alignment is determined between frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object

In some embodiments, determining that the temporal alignment between the one or more frames is present is based on temporal alignment results of one or more previous frames.

At operation 610, based on determining that the temporal alignment between the one or more frames is present, an intra-frame alignment scheme to spatially align charts within a frame may be determined.

In some embodiments, determining the intra-frame alignment may include spatially aligning the charts within the frame based on at least one of 3D location of respective charts, size of the respective charts, shape of the respective charts. In some embodiments, determining the intra-frame alignment may include spatially aligning texture charts within the frame based on geometry information associated with the charts and at least one of histogram of respective charts, mean RGB pixel values of the respective charts, or mean YUV pixel values of the respective charts.

At operation 615, the intra-frame alignment scheme may be applied to one or more corresponding charts in the one or more frames for inter-frame alignment.

In some embodiments, the one or more corresponding charts in the one or more frames may be charts that are similarly located across the one or more frames and have at least one of similar color values or similar shapes.

At operation 620, the one or more frames may be processed in parallel based on the intra-frame alignment and inter-frame alignment.

In some embodiments, processing the one or more frames in parallel may include dividing the one or more frames into hierarchical subgroups for parallel processing, wherein the hierarchical subgroups on a same level may be processed in parallel, and wherein higher level frames may only refer to lower level frames for the temporal alignment results.

In some embodiments, processing the one or more frames in parallel may include a first frame being processed independently of the one or more frames, and the one or more frames are processed in parallel based on the first frame.

In some embodiments, processing the one or more frames in parallel may include partitioning the one or more frames into one or more group of pictures (GoP), and processing each GoP among the one or more GoP in parallel. In some embodiments, the one or more frames may be partitioned into one or more hierarchical GoP structures.

The techniques, described above, can be implemented as computer software using computer-readable instructions and physically stored in one or more computer-readable media. For example, FIG. 7 shows a computer system 700 suitable for implementing certain embodiments of the disclosure.

The computer software can be coded using any suitable machine code or computer language, that may be subject to assembly, compilation, linking, or like mechanisms to create code including instructions that can be executed directly, or through interpretation, micro-code execution, and the like, by computer central processing units (CPUs), Graphics Processing Units (GPUs), and the like.

The instructions can be executed on various types of computers or components thereof, including, for example, personal computers, tablet computers, servers, smartphones, gaming devices, internet of things devices, and the like.

The components shown in FIG. 7 for computer system 700 are examples and are not intended to suggest any limitation as to the scope of use or functionality of the computer software implementing embodiments of the present disclosure. Neither should the configuration of components be interpreted as having any dependency or requirement relating to any one or combination of components illustrated in the non-limiting embodiment of a computer system 700.

Computer system 700 may include certain human interface input devices. Such a human interface input device may be responsive to input by one or more human users through, for example, tactile input (such as: keystrokes, swipes, data glove movements), audio input (such as: voice, clapping), visual input (such as: gestures), olfactory input (not depicted). The human interface devices can also be used to capture certain media not necessarily directly related to conscious input by a human, such as audio (such as: speech, music, ambient sound), images (such as: scanned images, photographic images obtain from a still image camera), video (such as two-dimensional video, three-dimensional video including stereoscopic video).

Input human interface devices may include one or more of (only one of each depicted): keyboard 701, mouse 702, trackpad 703, touch screen 710, data-glove, joystick 705, microphone 706, scanner 707, camera 708.

Computer system 700 may also include certain human interface output devices. Such human interface output devices may be stimulating the senses of one or more human users through, for example, tactile output, sound, light, and smell/taste. Such human interface output devices may include tactile output devices (for example tactile feedback by the touch-screen 710, data glove, or joystick 705, but there can also be tactile feedback devices that do not serve as input devices). For example, such devices may be audio

output devices (such as: speakers 709, headphones (not depicted)), visual output devices (such as screens 710 to include CRT screens, LCD screens, plasma screens, OLED screens, each with or without touch-screen input capability, each with or without tactile feedback capability— some of which may be capable to output two dimensional visual output or more than three dimensional output through means such as stereographic output; virtual-reality glasses (not depicted), holographic displays and smoke tanks (not depicted)), and printers (not depicted).

Computer system 700 can also include human accessible storage devices and their associated media such as optical media including CD/DVD ROM/RW 720 with CD/DVD or the like media 721, thumb-drive 722, removable hard drive or solid state drive 723, legacy magnetic media such as tape and floppy disc (not depicted), specialized ROM/ASIC/PLD based devices such as security dongles (not depicted), and the like.

Those skilled in the art should also understand that term “computer readable media” as used in connection with the presently disclosed subject matter does not encompass transmission media, carrier waves, or other transitory signals.

Computer system 700 can also include interface to one or more communication networks. Networks can for example be wireless, wireline, optical. Networks can further be local, wide-area, metropolitan, vehicular and industrial, real-time, delay-tolerant, and so on. Examples of networks include local area networks such as Ethernet, wireless LANs, cellular networks to include GSM, 3G, 4G, 5G, LTE and the like, TV wireline or wireless wide area digital networks to include cable TV, satellite TV, and terrestrial broadcast TV, vehicular and industrial to include CANBus, and so forth. Certain networks commonly require external network interface adapters that attached to certain general purpose data ports or peripheral buses 749 (such as, for example USB ports of the computer system 700; others are commonly integrated into the core of the computer system 700 by attachment to a system bus as described below (for example Ethernet interface into a PC computer system or cellular network interface into a smartphone computer system). Using any of these networks, computer system 700 can communicate with other entities. Such communication can be uni-directional, receive only (for example, broadcast TV), uni-directional send-only (for example CANbus to certain CANbus devices), or bi-directional, for example to other computer systems using local or wide area digital networks. Such communication can include communication to a cloud computing environment 755. Certain protocols and protocol stacks can be used on each of those networks and network interfaces as described above.

Aforementioned human interface devices, human-accessible storage devices, and network interfaces 754 can be attached to a core 740 of the computer system 700.

The core 740 can include one or more Central Processing Units (CPU) 741, Graphics Processing Units (GPU) 742, specialized programmable processing units in the form of Field Programmable Gate Areas (FPGA) 743, hardware accelerators for certain tasks 744, and so forth. These devices, along with Read-only memory (ROM) 745, Random-access memory 746, internal mass storage such as internal non-user accessible hard drives, SSDs, and the like 747, may be connected through a system bus 748. In some computer systems, the system bus 748 can be accessible in the form of one or more physical plugs to enable extensions by additional CPUs, GPU, and the like. The peripheral devices can be attached either directly to the core's system bus 748, or through a peripheral bus 749. Architectures for

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a peripheral bus include PCI, USB, and the like. A graphics adapter **750** may be included in the core **740**.

CPUs **741**, GPUs **742**, FPGAs **743**, and accelerators **744** can execute certain instructions that, in combination, can make up the aforementioned computer code. That computer code can be stored in ROM **745** or RAM **746**. Transitional data can be also be stored in RAM **746**, whereas permanent data can be stored for example, in the internal mass storage **747**. Fast storage and retrieve to any of the memory devices can be enabled through the use of cache memory, that can be closely associated with one or more CPU **741**, GPU **742**, mass storage **747**, ROM **745**, RAM **746**, and the like.

The computer readable media can have computer code thereon for performing various computer-implemented operations. The media and computer code can be those specially designed and constructed for the purposes of the present disclosure, or they can be of the kind well known and available to those having skill in the computer software arts.

As an example and not by way of limitation, a computer system having the architecture of computer system **700**, and specifically the core **740** can provide functionality as a result of processor(s) (including CPUs, GPUs, FPGA, accelerators, and the like) executing software embodied in one or more tangible, computer-readable media. Such computer-readable media can be media associated with user-accessible mass storage as introduced above, as well as certain storage of the core **740** that are of non-transitory nature, such as core-internal mass storage **747** or ROM **745**. The software implementing various embodiments of the present disclosure can be stored in such devices and executed by core **740**. A computer-readable medium can include one or more memory devices or chips, according to particular needs. The software can cause the core **740** and specifically the processors therein (including CPU, GPU, FPGA, and the like) to execute particular processes or particular parts of particular processes described herein, including defining data structures stored in RAM **746** and modifying such data structures according to the processes defined by the software. In addition or as an alternative, the computer system can provide functionality as a result of logic hardwired or otherwise embodied in a circuit (for example: accelerator **744**), which can operate in place of or together with software to execute particular processes or particular parts of particular processes described herein. Reference to software can encompass logic, and vice versa, where appropriate. Reference to a computer-readable media can encompass a circuit (such as an integrated circuit (IC)) storing software for execution, a circuit embodying logic for execution, or both, where appropriate. The present disclosure encompasses any suitable combination of hardware and software.

While this disclosure has described several non-limiting embodiments, there are alterations, permutations, and various substitute equivalents, which fall within the scope of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise numerous systems and methods which, although not explicitly shown or described herein, embody the principles of the disclosure and are thus within the spirit and scope thereof.

What is claimed is:

1. A method for parallel dynamic mesh alignment, the method being executed by at least one processor, the method comprising:

determining that a temporal alignment is present between one or more frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object based on temporal alignment results of one or more previous frames;

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based on determining that the temporal alignment between the one or more frames is present, determining an intra-frame alignment scheme to spatially align charts within a frame;

applying the intra-frame alignment scheme to one or more corresponding charts in the one or more frames for inter-frame alignment; and

processing the one or more frames in parallel based on the intra-frame alignment scheme and the inter-frame alignment, the processing comprising dividing the one or more frames into hierarchical subgroups for parallel processing, wherein the hierarchical subgroups on a same level are processed in parallel, and wherein higher level frames only refer to lower level frames for the temporal alignment results.

2. The method of claim 1, wherein the one or more corresponding charts in the one or more frames are charts that are similarly located across the one or more frames and have at least one of similar color values or similar shapes.

3. The method of claim 1, wherein determining the intra-frame alignment scheme comprises:

spatially aligning the charts within the frame based on at least one of a three-dimensional (3D) location of respective charts, a size of the respective charts, and a shape of the respective charts.

4. The method of claim 1, wherein determining the intra-frame alignment scheme comprises:

spatially aligning texture charts within the frame based on geometry information associated with the charts and at least one of histogram of respective charts, mean RGB pixel values of the respective charts, or mean YUV pixel values of the respective charts.

5. The method of claim 1, wherein a first frame is processed independently of the one or more frames, and the one or more frames are processed in parallel based on the first frame.

6. The method of claim 1, wherein the processing of the one or more frames in parallel comprises:

partitioning the one or more frames into one or more group of pictures (GoP); and
processing each GoP among the one or more GoP in parallel.

7. The method of claim 6, wherein the one or more frames are partitioned into one or more hierarchical GoP.

8. A device for parallel dynamic mesh alignment, the device comprising:

at least one memory configured to store program code; and

at least one processor configured to read the program code and operate as instructed by the program code, the program code including:

first determining code configured to cause the at least one processor to determine that a temporal alignment is present between one or more frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object based on temporal alignment results of one or more previous frames;

based on determining that the temporal alignment between the one or more frames is present, second determining code configured to cause the at least one processor to determine an intra-frame alignment scheme to spatially align charts within a frame;

first applying code configured to cause the at least one processor to apply the intra-frame alignment scheme to one or more corresponding charts in the one or more frames for inter-frame alignment; and

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first processing code configured to cause the at least one processor to process the one or more frames in parallel based on the intra-frame alignment scheme and the inter-frame alignment, the processing comprising dividing the one or more frames into hierarchical subgroups for parallel processing, wherein the hierarchical subgroups on a same level are processed in parallel, and wherein higher level frames only refer to lower level frames for the temporal alignment results.

9. The device of claim 8, wherein the one or more corresponding charts in the one or more frames are charts that are similarly located across the one or more frames and have at least one of similar color values or similar shapes.

10. The device of claim 8, wherein the first determining code comprises:

first spatially aligning code configured to cause the at least one processor to spatially align the charts within the frame based on at least one of a three-dimensional (3D) location of respective charts, a size of the respective charts, and a shape of the respective charts.

11. The device of claim 8, wherein the first determining code comprises:

first spatially aligning code configured to cause the at least one processor to spatially align texture charts within the frame based on geometry information associated with the charts and at least one of histogram of respective charts, mean RGB pixel values of the respective charts, or mean YUV pixel values of the respective charts.

12. The device of claim 8, wherein a first frame is processed independently of the one or more frames, and the one or more frames are processed in parallel based on the first frame.

13. The device of claim 8, wherein the processing code comprises:

first partitioning code configured to cause the at least one processor to partition the one or more frames into one or more group of pictures (GoP); and

second processing code configured to cause the at least one processor to process each GoP among the one or more GoP in parallel.

14. A non-transitory computer-readable medium storing instructions, the instructions comprising: one or more instructions that, when executed by one or more processors of a device for parallel dynamic mesh alignment, cause the one or more processors to:

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determine that a temporal alignment is present between one or more frames in a received input mesh including a plurality of polygons that describe a surface of a volumetric object based on temporal alignment results of one or more previous frames;

based on determining that the temporal alignment between the one or more frames is present, determine an intra-frame alignment scheme to spatially align charts within a frame;

apply the intra-frame alignment scheme to one or more corresponding charts in the one or more frames for inter-frame alignment; and

process the one or more frames in parallel based on the intra-frame alignment scheme and the inter-frame alignment, the processing comprising dividing the one or more frames into hierarchical subgroups for parallel processing, wherein the hierarchical subgroups on a same level are processed in parallel, and wherein higher level frames only refer to lower level frames for the temporal alignment results.

15. The non-transitory computer-readable medium of claim 14, wherein determining the intra-frame alignment scheme comprises:

spatially aligning the charts within the frame based on at least one of a three-dimensional (3D) location of respective charts, a size of the respective charts, and a shape of the respective charts.

16. The non-transitory computer-readable medium of claim 14, wherein determining the intra-frame alignment scheme comprises:

spatially aligning texture charts within the frame based on geometry information associated with the charts and at least one of histogram of respective charts, mean RGB pixel values of the respective charts, or mean YUV pixel values of the respective charts.

17. The non-transitory computer-readable medium of claim 14, wherein the processing of the one or more frames in parallel comprises:

partitioning the one or more frames into one or more group of pictures (GoP); and

processing each GoP among the one or more GoP in parallel.

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